

Welcome to SHM जैसे इसका नाम simple है वैसे ही Mains/NEET में इससे formula oriented simple question ही पूछे गए हैं but Adv. में WPE + Collision + Torque जैसे mix concept पूछे जाते हैं और ये chapter ऐसा है जिसके question आपको fluid, electrostatics, EMI, Magnetism etc. में भी देखने को मिलेंगे।



PERIODIC MOTION

- ★ Any motion which repeats itself after regular interval of time. The const. interval of time after which the motion is repeated is called Time Period.
Ex: Spring balance system, motion of earth around sun uniform circular motion, pendulum.

Oscillatory Motion

- ★ Motion is oscillatory if it moves to and fro about a fixed point after regular interval of time.

SHM

- ★ It is a special kind of oscillatory motion in which net force on particle at any time (any जगह) is proportional to negative of displacement from mean position.

$$\vec{F}_{\text{net}} \propto -\vec{x}$$

Net ext. force $\leftarrow \vec{F}_{\text{net}} = -k\vec{x}$

SHM Constant $\quad$ disp. of particle from mean position

Mean position (MP) बोले तो equilibrium position जहाँ पर $F_{\text{net}} = 0$ अबे भूल तो नहीं गए इसे NLM में पढ़ा था।



Eqⁿ of SHM

$$\vec{F}_{\text{net}} = -k\vec{x}$$

$$\vec{a} = -\frac{k}{m}\vec{x}$$

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x$$

Solve above equation & get

$$x = A \sin(\omega t + \phi)$$

disp. $\leftarrow$ $\rightarrow$ Amplitude

Where

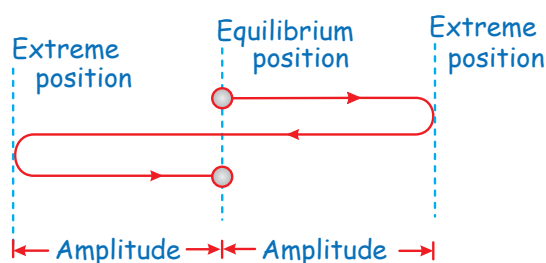
$$\omega = \sqrt{\frac{k}{m}} = \frac{2\pi}{T} \rightarrow \text{Time Period}$$

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$\rightarrow$ Time Period of SHM

- ★ $\frac{k}{m} = \omega^2 \Rightarrow k = m\omega^2$
- ★ $\omega = \text{Angular frequency} = \frac{2\pi}{T} = 2\pi f$
- ★ $A \rightarrow \text{Amplitude}$
- ★ $(\omega t + \phi) \rightarrow \text{phase}$
- ★ $\phi \rightarrow \text{Initial phase}$

If path of the particle performing SHM is



1. **Displacement:** The position of the particle with respect to the mean position at that instant. Displacement in SHM at time t is given by $x = A \sin(\omega t + \phi)$

2. **Amplitude (A):** It is the maximum value of displacement of the particle from its equilibrium position or it is a gap between mean position and extreme position.

Amplitude = $1/2$ (distance between extreme points/position).

3. **Time period (T):** Smallest time interval after which the oscillatory motion gets repeated is called time period.

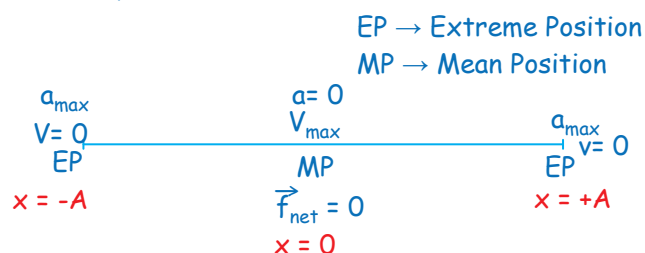
4. **Angular Frequency (ω):** $\omega = \frac{2\pi}{T}$ and its unit is rad/sec.

5. **Frequency (f):** Number of oscillations completed in unit time interval is called frequency of oscillations;

$$f = \frac{1}{T} = \frac{\omega}{2\pi} \text{ Its unit is } \text{sec}^{-1} \text{ or Hz.}$$

6. **Phase:** The physical quantity which represents the state of motion of particle (e.g. its position and direction of motion at any instant). Argument of sine function is called phase.

7. **Phase constant (ϕ):** Constant ϕ in equation of SHM is called phase constant or initial phase. It depends on initial position and direction of velocity.



★ $\vec{F}_{\text{net}}$ is always towards M.P.

★ Acc is always towards M.P.

★ Particle will speed up when move towards M.P. [chori].

★ Particle will speed down when move away from M.P.

SKC

अगर कभी SHM में particle के speed-up या speed-down में confusion हो तो mean position पर एक खूबसूरत छोरी या handsome छोरा (in case of girls) assume करो.... जब भी कभी तुम इसकी तरफ भागोगे speed-up होते जाओगे और जब इससे दूर भागोगे speed-down होते जाओगे.... तेरी real life का तो पता नहीं पर यहाँ तो assume कर ही ले....



Q. For a particle performing SHM, equation of motion is given as $\frac{d^2x}{dt^2} + 4x = 0$. Find the time period.

Sol. $\frac{d^2x}{dt^2} = -4x \Rightarrow \omega^2 = 4 \Rightarrow \omega = 2$

Time period: $T = \frac{2\pi}{\omega} = \pi$

Q. A particle is performing SHM its displacement from M.P. at any time is given as $x = 10 \sin(4\pi t + \pi/6)$. Answer following parts.

Sol. Comparing with $x = A \sin(\omega t + \phi)$

★ Amplitude = 10, $x_{\text{max}} = 10$

★ Angular frequency (ω) = 4π

★ Time period (T):

$$\therefore \omega = 2\pi/T = 4\pi$$

$$T = 1/2 \text{ sec}$$

★ Phase = $4\pi t + \pi/6$

★ Phase at $t = 1/4 \text{ sec}$

$$\text{Phase} = 4\pi \left(\frac{1}{4} \right) + \pi/6 = 7\pi/6.$$

★ Phase at $t = 0$

$$\text{Phase} = \pi/6 \text{ [Initial phase].}$$

★ Find disp of particle from M.P. at $t = 0$

$$x = 10 \sin(4\pi t + \pi/6)$$

$$\Rightarrow x = 10 \sin(0 + \pi/6) = 5$$

★ Find disp. of particle at $t = 1/4$ sec.

$$x = 10 \sin \left(4\pi \left(\frac{1}{4} \right) + \frac{\pi}{6} \right) = 10 \sin (180^\circ + 30^\circ)$$

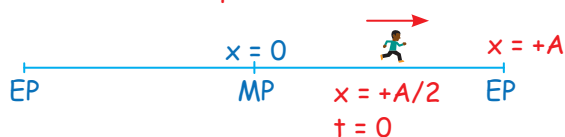
$$x = -5$$

★ Find velocity and acc of particle

$$v = \frac{dx}{dt} = 40\pi \cos \left(4\pi t + \frac{\pi}{6} \right)$$

$$a = \frac{dv}{dt} = -160\pi^2 \sin \left(4\pi t + \frac{\pi}{6} \right)$$

Q. A particle is performing SHM having amplitude A , angular frequency. ω on straight line at $t = 0$ particle is at $x = +A/2$ moving away from M.P. as shown. Find equⁿ of SHM.



Sol. $x = A \sin (\omega t + \phi)$

$$\phi = ?$$

$$t = 0, x = A/2$$

$$\Rightarrow \frac{A}{2} = A \sin(0 + \phi)$$

$$\sin \phi = 1/2$$

$$\phi = 30^\circ \text{ or } 150^\circ$$

$$\text{at } t = 0, v > 0$$

$$V = \frac{dx}{dt} = A\omega \cos(\omega t + \phi) > 0$$

$$V = A\omega \cos(0 + \phi) > 0$$

$$\cos \phi > 0$$

$$\therefore \phi = 30^\circ \text{ Hence eqn. of SHM will be}$$

$$x = A \sin (\omega t + 30^\circ)$$

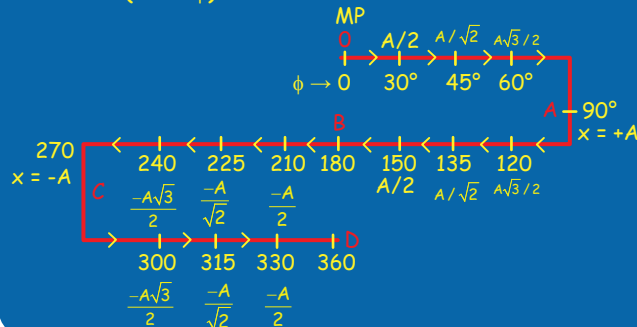
अगर $t = 0$ पर SHM करने वाले particle की location दे रखी तो अब हम नीचे दिया गया SKC method लगाएंगे बस तुम्हे ये मेहबारी करनी है की हर जगह पर लिखा हुआ angle याद करलो.... इससे आपका समय बचेगा नहीं तो mathematically या phasor से भी कर सकते हैं। Seriously बोल रहा हूँ इसे याद कर लेना वरना book से निकल कर मुक्का मारने आजाऊँगा



SKC

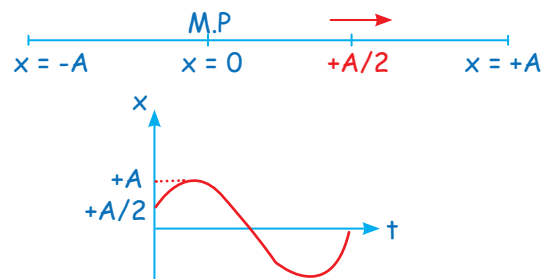
SKC-Method for phasor

$$x = A \sin (\omega t + \phi)$$

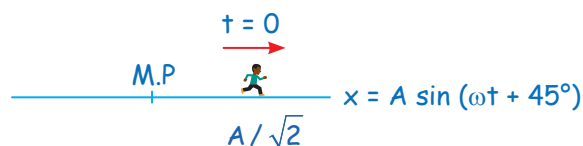


Q. A particle is performing SHM with A, ω . If at $t = 0$ particle is at $x = A/2$ moving away from M.P. find equⁿ of SHM

Sol. $x = A \sin (\omega t + 30^\circ)$



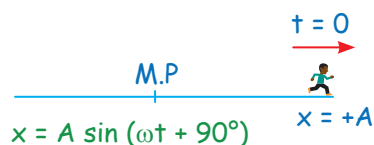
Q. A particle is performing SHM with A, ω . If at $t = 0$ particle is at $x = A/\sqrt{2}$ moving away from M.P. find equⁿ of SHM.

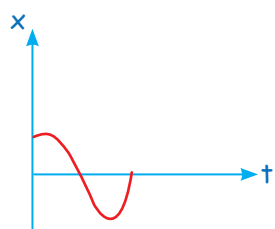


Q. A particle is performing SHM with A, ω . If at $t = 0$ particle is at $x = -A/2$ moving away from M.P. find equⁿ of SHM.

Sol. $x = A \sin (\omega t + 210^\circ)$

Q. A particle is performing SHM with A, ω . If at $t = 0$ particle is at $x = A$. Find equⁿ of SHM.





$$x = A \sin(\omega t + 90^\circ)$$

Q. A particle is performing SHM with A, ω . If at $t = 0$ particle is at $x = -A$. Find equⁿ of SHM.

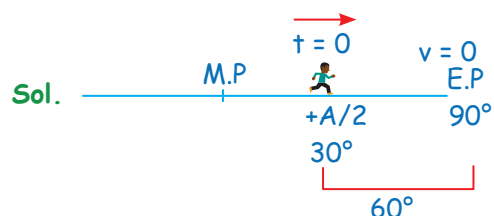
$$t = 0$$



$$x = A \sin(\omega t + 270^\circ)$$

Q. If at $t = 0$ particle is at $x = +A/2$ moving away from M.P.

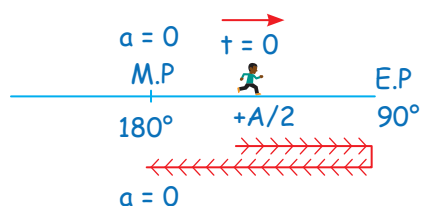
1. Find the time when particle comes to rest for the first time (reaches at EP)
2. Find time when it passes mean position ($a = 0$) for the first time.



$$1. 360^\circ \rightarrow T$$

$$1 \rightarrow T/360^\circ$$

$$60^\circ \rightarrow \frac{T}{360} \times 60 = \frac{T}{6}$$



$$2. 360^\circ \rightarrow T$$

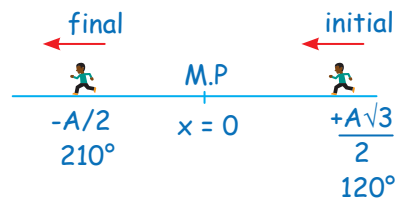
$$1^\circ \rightarrow T/360^\circ$$

$$150^\circ \rightarrow \frac{T}{360} \times 150 = \frac{5T}{12}$$

Q. A particle is performing SHM such that if at

$t = 0, x = \frac{A\sqrt{3}}{2}$ moving towards M.P. Find when it will reach $x = -A/2$ moving away from M.P.

Sol.



Angle wise अगर हम difference देखे तो

$$210^\circ - 120^\circ = 90^\circ$$

$$360^\circ \rightarrow T$$

$$1 \rightarrow \frac{T}{360^\circ}$$

$$90^\circ \rightarrow \frac{T}{4}$$

देखा कितना मस्त तरीका है..... इसलिए मैं कह रहा था कि वो table याद करलो!



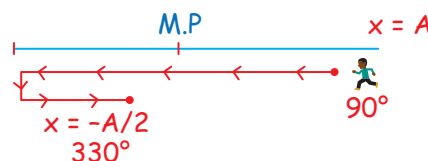
$$\rightarrow \text{Av. velocity} = \frac{x_f - x_i}{\text{time}} = \frac{(-A/2) - (A\sqrt{3}/2)}{T/4}$$

Q. Find min time taken by a particle performing SHM from initial ($x = +A$) to final, position

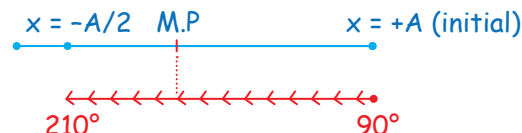
$\left(x = -\frac{A}{2}\right)$ moving towards MP.

Sol. Angle difference = $330^\circ - 90^\circ = 240^\circ$

$$t = \frac{T}{360} \times 240^\circ = \frac{2T}{3}$$



Q. Repeat the above problem if particle initially at $x = +A$ and finally $x = -A/2$ moving away from M.P.



Sol. Angle difference = $120^\circ, t = T/3$

Q. A particle performing SHM on straight line, at

$t = T/12$ particle is at $x = \frac{+A\sqrt{3}}{2}$ moving away from M.P.

Find equⁿ of SHM.

Sol. $x = A \sin(\omega t + 60^\circ - 30^\circ) = A \sin(\omega t + 30^\circ)$

अब थोड़ा सा kinematics खेलते हैं।

$$x = A \sin(\omega t + \phi)$$

$$v = A\omega \cos(\omega t + \phi) \rightarrow V_{\max} = A\omega \text{ (At M.P.)}$$

$$a = -A\omega^2 \sin(\omega t + \phi) \rightarrow a_{\max} = A\omega^2 \text{ (magnitude)}$$

$$a = -\omega^2 x$$

$$a = -\omega^2 x$$

$$V = \omega \sqrt{A^2 - x^2}$$

$$V_{\max} = A\omega \text{ (At M.P.)}$$

$$V_{\min} = 0, x = \pm A$$



Very Important Formula

इसे गलती से भी मत भुलना
क्योंकि ये हमें किसी जगह पर
speed बताएगा

Q. If eqn of SHM is $x = 10 \sin(2\pi t + \pi/6)$, Find

$V_{\max}$, $a_{\max}$ and speed at $x = 6$.

Sol. $V_{\max} = A\omega = 10 \times 2\pi = 20\pi$

$$a_{\max} = A\omega^2 = 10 \times (2\pi)^2 = 40\pi^2$$

Speed at $x = 6$

$$V = \omega \sqrt{A^2 - x^2} = 2\pi \sqrt{100 - 36} = 16\pi$$

SKC



Mean position se equal dist. pr + or-,
speed same hogi hamesha.

Q. A particle executing SHM (A, ω, T)

Find its speed at a dist. of half of amplitude
from M.P.

Sol. $x = \frac{A}{2}, V = \omega \sqrt{A^2 - \left(\frac{A}{2}\right)^2} = A\omega \frac{\sqrt{3}}{2} = V_{\max} \times \frac{\sqrt{3}}{2}$

Q. At what distance from M.P. speed of particle
will be half of max speed.

Sol. $V = \omega \sqrt{A^2 - x^2}$

$$\frac{A\omega}{2} = \omega \sqrt{A^2 - x^2}$$

$$x = \pm \frac{A\sqrt{3}}{2}$$

ENERGY ANALYSIS

$$v = \omega \sqrt{A^2 - x^2}$$

$$K.E. = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(A^2 - x^2)$$

$$K.E. = \frac{1}{2}K(A^2 - x^2)$$

$$K = m\omega^2$$

at, $x = 0, (K.E.)_{\max} = \frac{1}{2}KA^2$

Total mechanical energy:

$$T.M.E. = K.E. + P.E.$$

$$\int dU = -\int F dx$$

$$\int_{U_0}^U dU = -\int_0^x kx dx$$

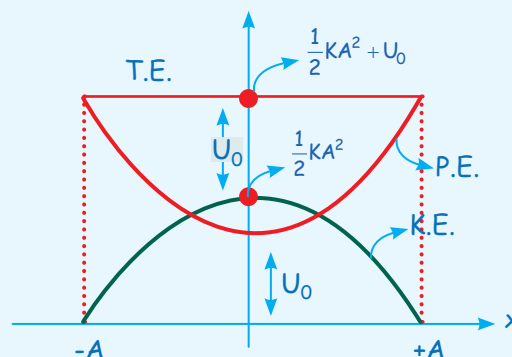
$$U - U_0 = -\frac{1}{2}Kx^2$$

$$U = \frac{1}{2}Kx^2 + U_0 \quad (U_0 = P.E. \text{ at M.P.})$$

$$K.E. = \frac{1}{2}K(A^2 - x^2)$$

$$P.E. = \frac{1}{2}Kx^2 + U_0$$

$$T.E. = \frac{1}{2}KA^2 + U_0 = \text{Const.}$$

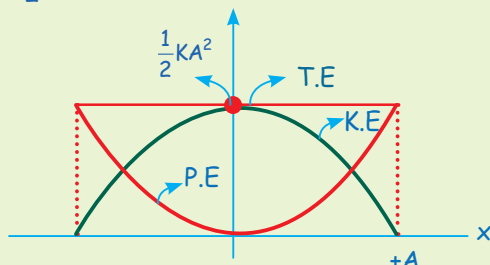


If $U_0 = 0$

$$P \cdot E = \frac{1}{2} Kx^2$$

$$K \cdot E = \frac{1}{2} K(A^2 - x^2)$$

$$T \cdot E = \frac{1}{2} KA^2 = \text{const.}$$



यार एक बार speed, kinetic energy, potential energy का variation time के according भी देखलो और खुद से rough copy पर practice करो।

Let $x = A \sin \omega t$

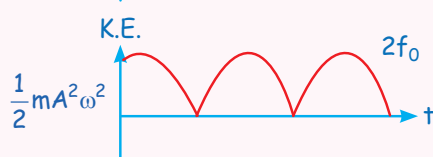
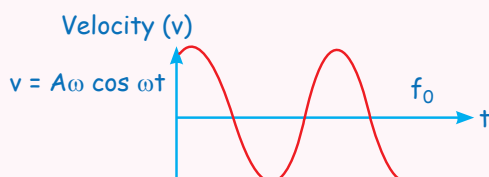
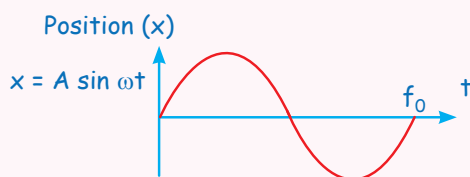
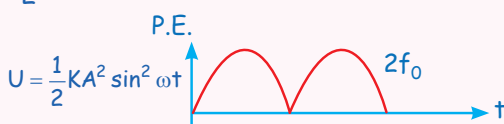
$$v = \frac{dx}{dt} = A\omega \cos \omega t$$

$$v = A\omega \cos \omega t$$

$$K \cdot E = \frac{1}{2} mv^2 = \frac{1}{2} mA^2 \omega^2 \cos^2 \omega t$$

$$P \cdot E = \frac{1}{2} Kx^2, (U_0 = 0)$$

$$P \cdot E = \frac{1}{2} KA^2 \sin^2 \omega t$$

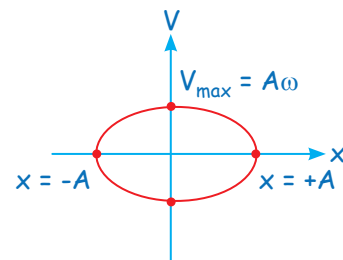


$$\star v = \omega \sqrt{A^2 - x^2}$$

$$\frac{v^2}{\omega^2} = A^2 - x^2$$

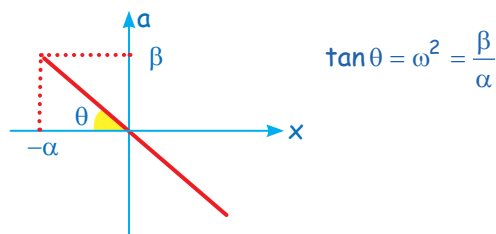
$$\frac{v^2}{\omega^2} + x^2 = A^2$$

$$\frac{v^2}{A^2 \omega^2} + \frac{x^2}{A^2} = 1 \quad \text{[Ellipse]}$$



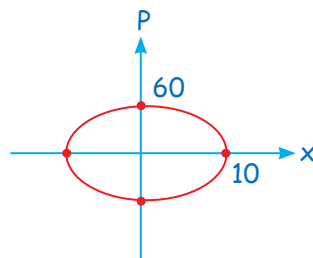
$$\star a = -\omega^2 x$$

[Graph of x vs. a = st. line]



Q. Find speed of particle when it is at a dist. of 8m from M.P. from momentum (P) displacement (x) graph.

Sol.



$$P_{\max} = mV_{\max}$$

$$\Rightarrow 60 = 2V_{\max}$$

$$\Rightarrow V_{\max} = 30 = A\omega$$

$$\omega = 3$$

$$V = 3\sqrt{100 - 64} = 18$$

Q. Total energy of particle executing SHM is E_0 , ($U_0 = 0$) What is K.E. When disp. is half of amplitude & PE.

Sol. $E_0 = \frac{1}{2} KA^2$ and $P \cdot E = \frac{1}{2} Kx^2 = \frac{1}{2} K \left(\frac{A}{2} \right)^2$

$$= \frac{1}{8} KA^2 = \frac{E_0}{4}$$

$$K \cdot E = E_0 - \frac{E_0}{4} = \frac{3E_0}{4}$$

Q. If $U_0 = 0$. Find location when K.E is equal to P.E.

Sol. $\frac{1}{2}K(A^2 - x^2) = \frac{1}{2}Kx^2$

$$A^2 = 2x^2$$

$$x = \pm A / \sqrt{2}$$

SHM equⁿ

Which of the following question can represent SHM.

★ $\vec{F}_{\text{net}} = -K\vec{x}$ ✓

★ $\vec{F}_{\text{net}} = -Kx^2$ ✗

★ $F_{\text{net}} = -10x$ ✓

★ $F_{\text{net}} = -10x + 20$ ✓

SKC: $x = a \sin \omega t + b \cos \omega t$
 $\cos \omega t \text{ AmP}$
 $= \sqrt{a^2 + b^2}$

Q. Find whether given eqn can represent SHM or not if yes find its amplitude

$$x = A \sin \omega t + A\sqrt{3} \cos \omega t$$

Sol. $x = 2A \left(\frac{1}{2} \sin \omega t + \frac{\sqrt{3}}{2} \cos \omega t \right)$

$$x = 2A (\sin \omega t \cdot \cos 60^\circ + \cos \omega t \cdot \sin 60^\circ)$$

$$x = 2A \sin (\omega t + 60^\circ)$$

Ans. Yes it is SHM with amplitude 2A

If eqn of SHM is $x = a \sin \omega t + b \cos \omega t$

$$\text{Amplitude} = \sqrt{a^2 + b^2}$$

SHM में time period निकालने के लिए नीचे 4 point लिख रहा हूँ। इन्हें अच्छे से याद कर लेना क्योंकि ये आगे बहुत काम आएंगे।

1. सबसे पहले mean position पता करो (Find out MP)
2. अब particle को MP से x displaced करके FBD बनाओ।

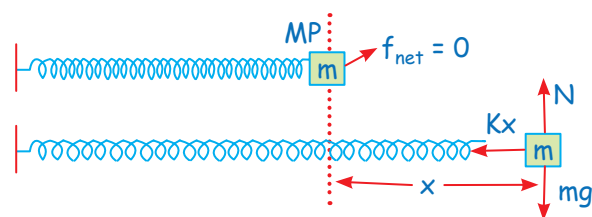
3. अब particle पर net force निकालो

4. अगर $\vec{F}_{\text{net}} = -K\vec{x}$ आया तो ये SHM है जिसका time period

$$T = 2\pi \sqrt{\frac{m}{k}} \text{ होगा}$$



SPRING BLOCK SYSTEM

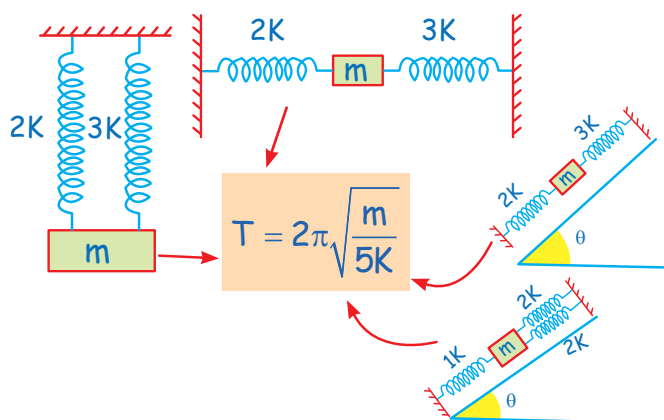


$$\vec{F}_{\text{net}} = \vec{F}_{\text{SP}} + \vec{F}_g + \vec{F}_N$$

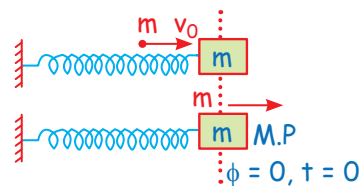
$$\vec{F}_{\text{net}} = Kx \text{ (magnitude)}$$

$$\vec{F}_{\text{net}} = -K\vec{x}$$

$$T = 2\pi \sqrt{\frac{m}{K}}$$



Q. A bullet of mass m strike and stick to the block at $t = 0$, of same mass which is connected to spring as shown in fig. If Spring is in natural length at $t = 0$. Find time period, amplitude of SHM.



Sol. इसकी M.P → NL पर होगी

Bullet strike & stick to the mass at $t = 0$

$$(P_i)_{\text{just before}} = (P_f)_{\text{just after}}$$

$$mv_0 + 0 = 2mv_f$$

$$v_f = \frac{v_0}{2}$$

★ Time Period-

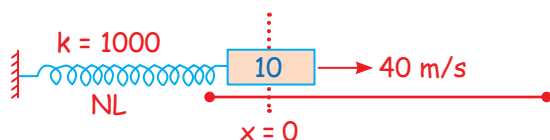
$$T = 2\pi\sqrt{\frac{2m}{k}}, \quad \omega = \sqrt{\frac{k}{2m}}$$

★ Amplitude-

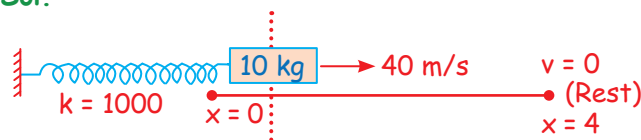
$$V_{\max} = A\omega$$

$$A = V_{\max}/\omega = \frac{v_0}{2} \sqrt{\frac{2m}{k}}$$

Q. A block is at rest at equilibrium on a horizontal floor as shown in fig. A sharp impulse is given to the block such that. At $t = 0^+$ block acquire velocity of $40 \text{ m/s } \hat{i}$. Find amplitude, $a_{\max}$ and eqn of SHM impulse given and Time period.



Sol.

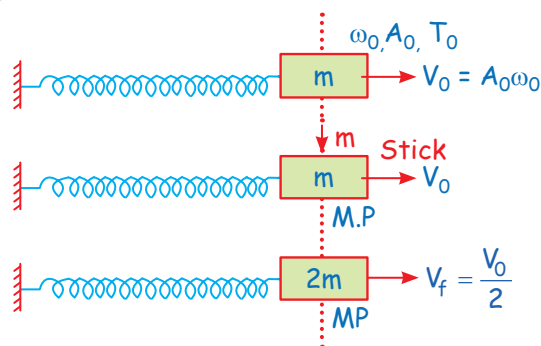


$$T = 2\pi\sqrt{\frac{10}{1000}} = \frac{\pi}{5}, \quad \omega = \sqrt{\frac{k}{m}} = 10$$

$$\begin{aligned} V_{\max} &= A\omega & a_{\max} &= A\omega^2 & \text{SHM} &= \\ 40 &= A \times 10 & &= 4 \times 100 & x &= A \sin(\omega t + 0) \\ A &= 4 \text{ m} & &= 400 \text{ ms}^{-2} & x &= 4 \sin(10t) \end{aligned}$$

Q. A spring block system of (m, K) is performing SHM (ω_0, A_0, T_0) on smooth horizontal surface. When block is crossing in MP a particle of same mass m is gently placed on the block such that particle stick to the mass. Find new amplitude new time period of new SHM.

Sol.



$$\omega_0 = \sqrt{\frac{k}{m}} = \omega_{\text{old}}$$

$$\omega_{\text{new}} = \frac{\omega_0}{\sqrt{2}}$$

$$P_i = P_f$$

$$mv_0 + 0 = 2mv_f$$

$$V_f = \frac{V_0}{2}$$

$$V_f = \frac{V_0}{2} = \frac{A_0\omega_0}{2}$$

$$A_{\text{new}}\omega_{\text{new}} = \frac{A_0\omega_0}{2}$$

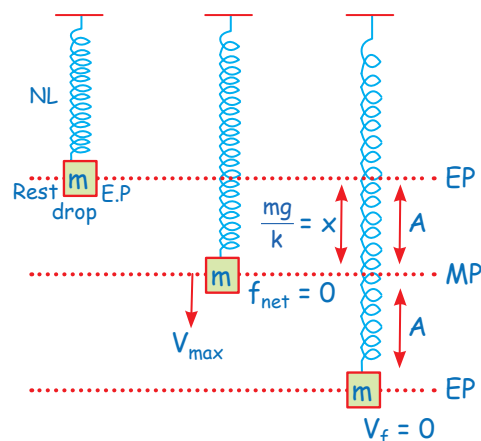
$$A_{\text{new}} \frac{\omega_0}{\sqrt{2}} = \frac{A_0\omega_0}{2}$$

$$A_{\text{new}} = \frac{A_0}{\sqrt{2}}$$

$$x = \frac{A_0}{\sqrt{2}} \sin\left(\sqrt{\frac{k}{2m}}t + 0\right)$$

Q. Mass is released from rest as shown in diagram from natural length (NL). Find time period, amplitude, max velocity of a particle.

Sol.



$$\text{Amplitude} = \frac{mg}{k}$$

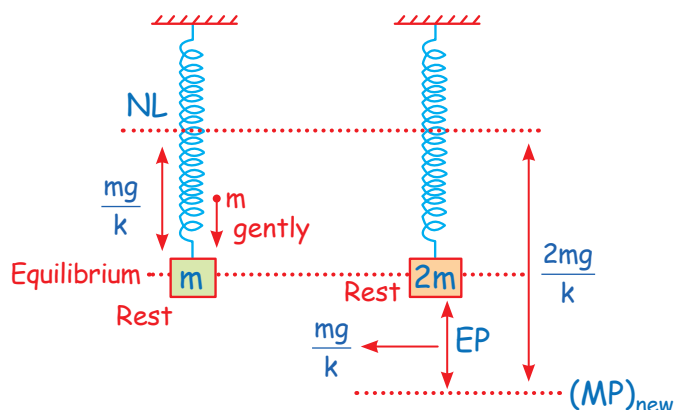
$$\text{Maximum velocity } V_{\max} = A\omega$$

$$= \frac{mg}{k} \cdot \sqrt{\frac{k}{m}}$$

$$= g\sqrt{\frac{m}{k}}$$

Q. In a spring block system, block is at rest and in equilibrium when it is suspended from the ceiling. At $t = 0$, a particle of mass m is gently dropped on block. Find A .

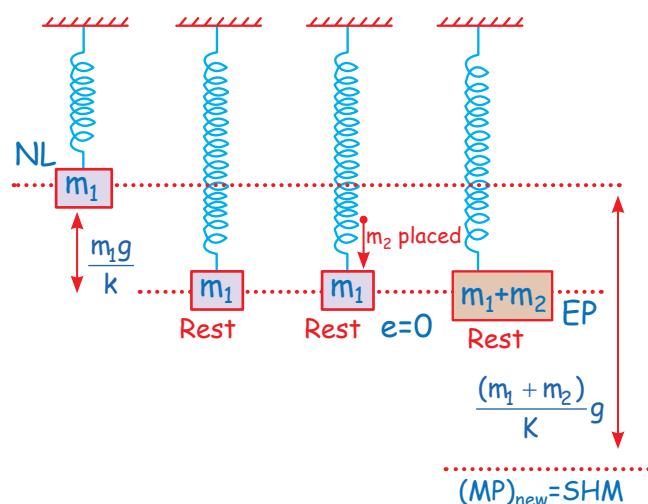
Sol.



$$A = \frac{mg}{k}$$

Q. A particle dropped on m_1 as shown. Find T , ω and A .

Sol.



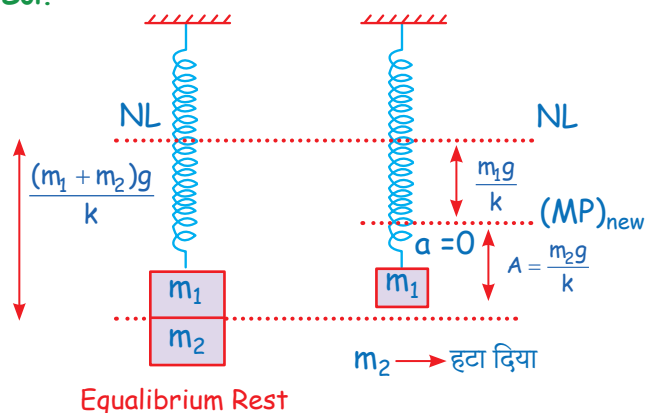
$$T = 2\pi \sqrt{\frac{m_1 + m_2}{k}}$$

$$\omega = \sqrt{\frac{k}{m_1 + m_2}}$$

$$A = \frac{m_2 g}{k}$$

Q. Two mass m_1 and m_2 are stick together and suspended from a ceiling by a spring such that they are at rest and in equilibrium. If from equilibrium m_2 is removed. Find T , ω and A .

Sol.



$$T = 2\pi \sqrt{\frac{m_1}{k}}, \quad \omega = \sqrt{\frac{k}{m_1}} \quad \text{and} \quad A = \frac{m_2 g}{k}$$

ANGULAR SHM

★ In angular SHM, the restoring torque (or acceleration) acting on the particle should always be proportional to the angular displacement of the particle and directed towards the equilibrium position.

$$\therefore \tau \propto -\theta \Rightarrow \alpha \propto -\theta.$$

जैसे पहले linear SHM में time period निकालना सीखा था same as it is यहाँ पर है बस force की जगह torque आ जाएगा displacement की जगह angular displacement mass के जगह moment of inertia and so on.... To find the time period of angular SHM

1. सबसे पहले mean position पता करो जहाँ torque = 0
2. अब particle को MP से θ displaced करके FBD बनाओ।
3. अब particle पर net torque निकालो।

4. अगर $\vec{\tau}_{net} = -C\vec{\theta}$ आया तो ये angular SHM है जिसका time period $T = 2\pi \sqrt{\frac{I}{C}}$ होगा।



m	$\longrightarrow$	I
x	$\longrightarrow$	θ
v	$\longrightarrow$	$\vec{\omega} = d\theta/dt$
$f = ma$	$\longrightarrow$	$\tau = I\alpha$
P	$\longrightarrow$	L
J	$\longrightarrow$	H

Linear SHM	Angular SHM
$\vec{F}_{\text{net}} = -K\vec{x}$	$\vec{\tau}_{\text{net}} = -K'\vec{\theta}$
$T = 2\pi\sqrt{\frac{m}{K}}, \omega = \sqrt{\frac{K}{m}}$	$T = 2\pi\sqrt{\frac{I}{K'}}, \omega = \frac{2\pi}{T} = \sqrt{\frac{K'}{I}}$
$x = A \sin(\omega t + \phi)$	$\theta = \theta_0 \sin(\omega t + \phi)$
$v = \omega\sqrt{A^2 - x^2}$	$\frac{d\theta}{dt} = \Omega, \Omega = \omega\sqrt{A^2 - x^2}$
$K.E = \frac{1}{2}K(A^2 - x^2)$	$K.E = \frac{1}{2}K(\theta_0^2 - \theta^2)$
$P.E = \frac{1}{2}KA^2 + u_0$	$P.E = \frac{1}{2}K\theta^2 + u_0$
$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$	$\frac{d^2\theta}{dt^2} + \frac{C}{I}\theta = 0$

COMPOUND PENDULUM

Consider a body hinged at O.

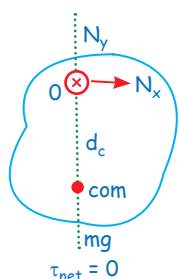


Fig → Equilibrium

$$\tau_0 = mg \sin\theta \times d_c$$

$$\tau_0 = mg d_c \sin\theta$$

If ' θ ' is small

$$\sin\theta \approx \theta$$

$$\tau = mg d_c \theta$$

$$\vec{\tau} = mg d_c \vec{\theta}$$

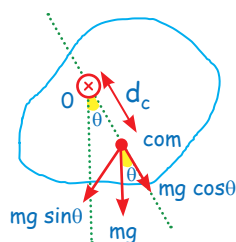


Fig → Displaced from Equilibrium

$$\vec{\tau}_{\text{net}} = -K\vec{\theta}$$

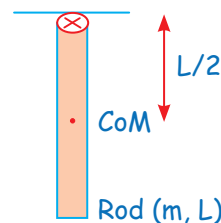
$$T = 2\pi\sqrt{\frac{I}{K}} = 2\pi\sqrt{\frac{I}{mgd_c}}$$

$$T = 2\pi\sqrt{\frac{I}{mgd_c}}$$

$I \rightarrow$ moI abt hinged point.

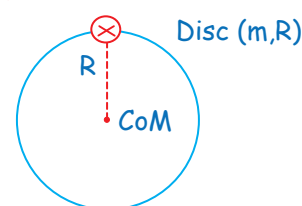
$d_c \rightarrow$ CoM ka hinged pt. se dist.

Q. A rod is hinged as shown in figure. Find time period of oscillation.



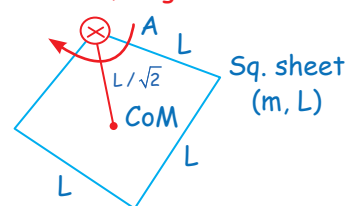
$$\text{Sol. } T = 2\pi\sqrt{\frac{I}{mgd_c}} \Rightarrow T = 2\pi\sqrt{\frac{mL^2/3}{mgL/2}} = 2\pi\sqrt{\frac{2I}{3g}}$$

Q. A ring hinged as shown in figure. Find time period of oscillation.



$$\text{Sol. } T = 2\pi\sqrt{\frac{I}{mgd_c}} = 2\pi\sqrt{\frac{mR^2/2 + mR^2}{mgR}}$$

Q. Find time period of oscillation for a square sheet of side L, hinged as shown in fig.

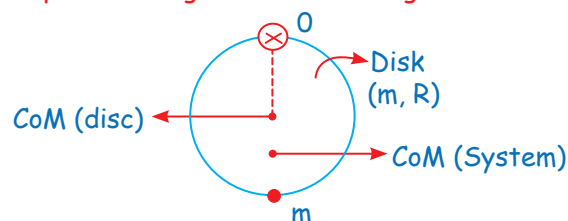


$$\text{Sol. } T = 2\pi\sqrt{\frac{I_A}{mgd_c}}$$

$$I_A = \frac{mL^2}{6} + m\left(\frac{L}{\sqrt{2}}\right)^2$$

$$d_c = L/\sqrt{2}$$

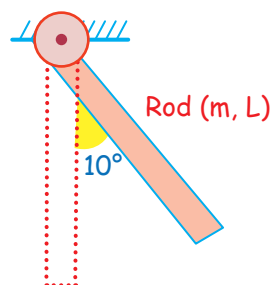
Q. Find time period of oscillation for disc and particle hinged as shown in figure.



$$\text{Sol. } T = 2\pi\sqrt{\frac{I_0}{(2m)g(R + R/2)}}$$

$$I_0 = \frac{mR^2}{2} + mR^2 + m(2R)^2$$

Q. Find time period of oscillation if a Rod (m, L) is dropped from small angular displacement.



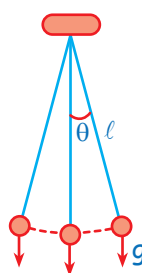
Sol. $T = 2\pi \sqrt{\frac{I}{mgd_c}} = 2\pi \sqrt{\frac{mL^2}{3mgL/2}}$
 $\omega = 2\pi/T$

SIMPLE PENDULUM

$$T = 2\pi \sqrt{\frac{l}{g}} = 2\pi \sqrt{\frac{ml^2}{mgl}}$$

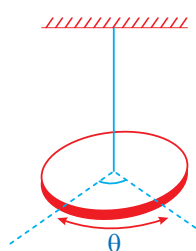
$$T = 2\pi \sqrt{\frac{l}{g}}$$

left $\rightarrow g = g_{\text{eff}}$.



Torsional Pendulum

In torsional pendulum, an extended object is suspended at the centre by a light torsion wire. A torsion wire is essentially inextensible, but is free to twist about its axis. When the lower end of the wire is rotated by a slight amount the wire applies a restoring torque causing the body to oscillate rotationally.



Let the twist given is θ , when released, the restoring torque produced.

$\tau = -C\theta$ where, C = Torsional constant or, $I\alpha = -C\theta$

Where, I = Moment of inertia about the vertical axis.

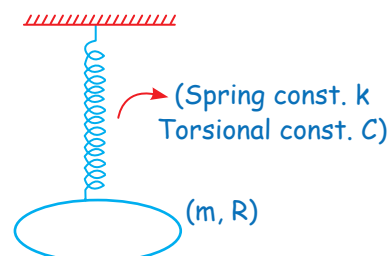
or, $\alpha = -\frac{C}{I} \theta$

$\therefore$ Time period, $T = 2\pi \sqrt{\frac{I}{C}}$

Q. In following figure find the ratio of time period of T_1 and T_2 if

$T_1 \rightarrow$ spring block system

$T_2 \rightarrow$ Torsional pendulum

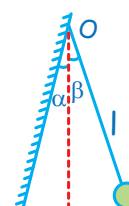


Sol. $T_1 = 2\pi \sqrt{\frac{m}{k}}$

$$T_2 = 2\pi \sqrt{\frac{I}{C}}$$

$$\frac{T_1}{T_2} = \sqrt{\frac{m}{k} \times \frac{C}{mR^2/2}}$$

Q. A ball is suspended by a thread of length l at the point O on the wall, forming a small angle α with the vertical as shown in figure. Then the thread with the ball was deviated through a small angle β ($\beta > \alpha$) and set free. Assuming the collision of the ball against the wall to be perfectly elastic, find the oscillation period of such a pendulum.



Sol. The system can be assumed as an incomplete SHM.

Thus time period can be calculate as twice the (time taken by the bob to travel to mean position from extreme + time taken to travel from mean to wall)

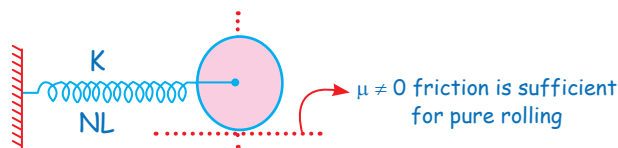
$$\therefore T_{\text{net}} = 2 \left[\frac{T}{4} + \frac{\sin^{-1}\left(\frac{\alpha}{\beta}\right)T}{2\pi} \right] = 2 \left[\frac{2\pi}{4} \cdot \sqrt{\frac{l}{g}} + \frac{\sin^{-1}\left(\frac{\alpha}{\beta}\right)}{2\pi} \cdot 2\pi \sqrt{\frac{l}{g}} \right]$$

$$T = 2\sqrt{l/g} [\pi/2 + \sin^{-1}(\alpha/\beta)]$$

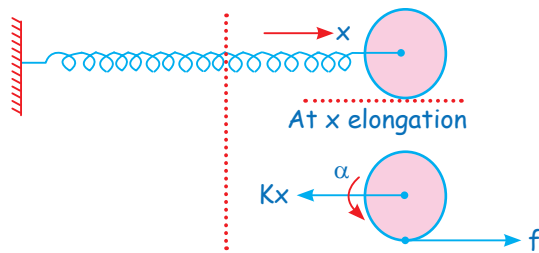
अब मैं SHM के तुम्हारे concept को और strong करने के लिए कुछ selected and important questions attach कर रहा हूँ इन्हें अच्छे से practice कर लेना.... अब सुन खुद से पहले try करना तभी फायदा है।



Q. A solid cylinder of mass M and radius R is attached to a spring of stiffness k as shown in the figure. The cylinder can roll without slipping on a rough horizontal surface. Show that the centre of mass of the cylinder executes SHM and determine its time period.



Sol. For small disp. find $T = ?$



$$Kx - f = ma \quad (a = R\alpha)$$

$$\tau = I\alpha$$

$$f \cdot R = I\alpha = \frac{mR^2}{2} \cdot \frac{a}{R}$$

$$f = \frac{ma}{2} \quad (1)$$

$$Kx - \frac{ma}{2} = ma$$

$$Kx = \frac{3}{2}ma \quad (2)$$

$$a = \frac{2K}{3m}x$$

$$a = -\frac{2K}{3m}x = -\omega^2 x$$

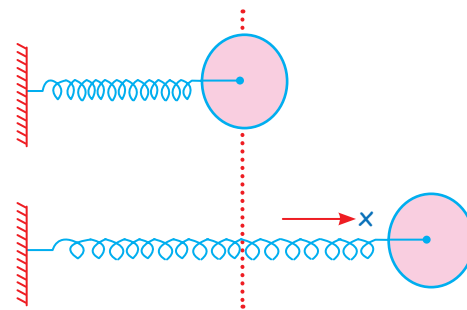
$$\omega^2 = \frac{2K}{3m} \Rightarrow \omega = \sqrt{\frac{2K}{3m}} = \frac{2\pi}{T}$$

$$T = 2\pi \sqrt{\frac{3m}{2K}}$$

ऐसे सवाल के लिए एक और method है.... अरे एक बात बता SHM में total energy constant है तो total energy का differentiation wrt time zero होगा ना जरा एक बार करके देख



Method - 2



T.E = Cost.

$$T.E = \frac{1}{2}Kx^2 + \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$T.E = \frac{1}{2}Kx^2 + \frac{1}{2}mv^2 + \frac{1}{2} \cdot \frac{mR^2}{2} \cdot \frac{V^2}{R^2} \quad (V = R\omega)$$

$$T.E = \frac{1}{2}Kx^2 + \frac{3}{4}mv^2$$

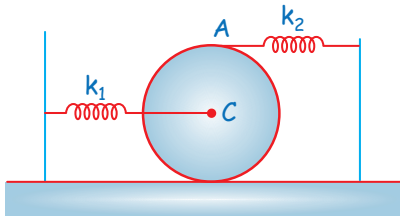
$$\frac{d(T.E)}{dt} = \frac{1}{2}K2x \cdot v + \frac{3}{4}m2v \cdot a = 0$$

$$Kx + \frac{3}{2}ma = 0$$

$$a = -\frac{2K}{3m}x$$

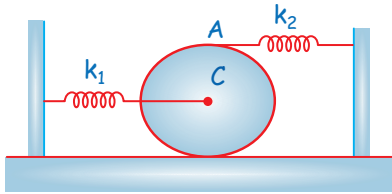
$$a = -\omega^2 x \Rightarrow T = 2\pi \sqrt{\frac{3m}{2K}}$$

Q. A solid cylinder of mass $m = 1$ kg is kept in equilibrium on a horizontal surface. Two unstretched springs of force constant $k_1 = 20$ N/m and $k_2 = 20$ N/m are attached to the cylinder shown in figure. Find the period of small oscillations. Cylinder rolls without sliding.



Sol. Let at any instant centre of cylinder is displaced by x (towards left).

Then spring attached at C is compressed by x and spring attached at A elongates by $2x$. Let v be the velocity of centre of cylinder and ω its angular velocity.



Total mechanical energy in displaced position is

$$E = \frac{1}{2}mv^2 + \frac{1}{2}I_C\omega^2 + \frac{1}{2}k_1x^2 + \frac{1}{2}k_2(2x)^2$$

But $\omega = \frac{v}{R}$ and $I_C = \frac{1}{2}mR^2$

Hence, $E = \frac{3}{4}mv^2 + \frac{1}{2}k_1x^2 + 2k_2x^2$

In case of pure rolling energy is conserved

$$\therefore \frac{dE}{dt} = 0 \text{ or } \frac{3}{2}mv \left(\frac{dv}{dt} \right) + k_1x \left(\frac{dx}{dt} \right) + 4k_2x \left(\frac{dx}{dt} \right) = 0$$

$\frac{dx}{dt} = v$ and $\frac{dv}{dt} = a$ (acceleration), with these

Substitutions we get, $\frac{3}{2}ma = -(k_1 + 4k_2)x$

Since $a \propto -x$ oscillations are simple harmonic in nature.

Time period of which is given by

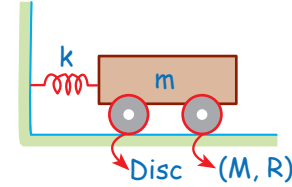
$$T = 2\pi \sqrt{\frac{x}{a}} = 2\pi \sqrt{\frac{3m}{2(k_1 + 4k_2)}}$$

Substituting the values we get

$$T = 2\pi \sqrt{\frac{3 \times 1}{2(10 + 4 \times 20)}} \text{ or } T = 0.81 \text{ s}$$

Q. A cart consists of a body and four wheels on frictionless axles. The body has a mass m . The wheels are uniform discs of mass M and radius R . The cart rolls, without slipping, back and

forth on a horizontal plane under the influence of a spring attached to one end of the cart (figure). The spring constant is k . Taking into account the moment of inertia of the wheels, find a formula for the frequency of the back and forth motion of the cart.



Sol. $\omega = \sqrt{\frac{k}{(m + 6M)}}$

$$E = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 + 4 \left[\frac{1}{2}Mv^2 + \frac{1}{2} \left(\frac{MR}{2} \right)^2 \frac{V^2}{R^2} \right] = \text{const}$$

$$E = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 + 3Mv^2$$

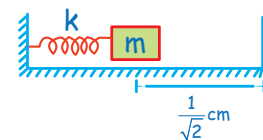
$$\frac{dE}{dt} = kx \frac{dx}{dt} + mv \frac{dv}{dt} + 6Mv \frac{dv}{dt} = 0$$

$$\Rightarrow kx = -(m + 6M)(-\omega^2 x)$$

$$\Rightarrow \omega = \sqrt{\frac{k}{m + 6M}}$$

$$\therefore f = \frac{1}{2\pi} \sqrt{\frac{k}{m + 6M}}$$

Q. A block of mass 0.9 kg attached to a spring of force constant k is lying on a frictionless floor. The spring is compressed to $\sqrt{2}$ cm and the block is at a distance $\frac{1}{\sqrt{2}}$ cm from the wall as shown in the figure. When the block is released, it makes elastic collision with the wall and its period of motion is 0.2 sec. Find the approximate value of k .

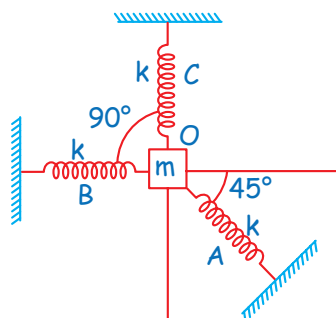


Sol. $T = \frac{2\pi}{3} \sqrt{\frac{0.9}{k}} = 0.2$ (given)

$$\frac{4 \times 10}{9} \times \frac{9}{10 \times k} = \frac{4}{100}$$

$$\Rightarrow k \approx 100 \text{ N/m}$$

- Q.** A particle of mass m is attached to three identical springs A, B and C each of force constant k as shown in figure. If the particle of mass m is pushed slightly against the spring A and released, find the time period of oscillations.



Sol. When the particle of mass m at O is pushed by y in the direction of A , spring A will be compressed by y while B and C will be stretched by $y' = y \cos 45^\circ$, so the total restoring force on the mass m along OA is given by

$$F = -[F_A + F_B \cos 45^\circ + F_C \cos 45^\circ]$$

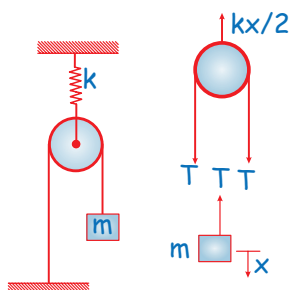
$$F = -[ky + 2(ky') \cos 45^\circ]$$

$$F = -[ky + 2k(y \cos 45^\circ) \cos 45^\circ]$$

$$\text{i.e., } F = -k'y \quad \text{where } k' = 2k$$

$$\text{so } T = 2\pi \sqrt{\frac{m}{k'}} = 2\pi \sqrt{\frac{m}{2k}}$$

- Q.** For the arrangement shown in the figure, find the period of oscillation.



Sol. Obviously, when the block is displaced down by x , the spring will stretch by $\frac{x}{2}$.

From the free body diagram of the pulley,

$$\frac{kx}{2} = 2T \quad \text{or } T = \frac{kx}{4}$$

The net restoring force on the block is T .

Using the Second Law of motion, we get

$$m \frac{d^2x}{dt^2} = -T = -\frac{kx}{4}$$

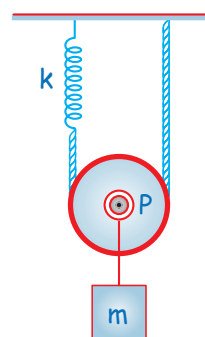
$$\text{or } \frac{d^2x}{dt^2} + \left(\frac{k}{4m}\right)x = 0$$

Thus, the period of SHM is given by

$$T = 2\pi \sqrt{\frac{4m}{k}}$$

Note that we have not taken gravity into account as it does not affect the time period.

- Q.** Find the time period of oscillation of m if pulley P is light and small. (Rope is massless)



Sol. $T = kx_0$

Initially, $2T = 2kx_0$.

$$\therefore 2kx_0 = mg$$

Let the displacement of block at any time is x and stretch of spring is x' , then

$$\frac{x' + 0}{2} = x \Rightarrow x' = 2x$$

Energy of the system is

$$\frac{1}{2}mv^2 + \frac{1}{2}k(x_0 + x')^2 - mgx = \text{constant}$$

Differentiating w.r.t. time

$$mv \frac{dv}{dt} + k(x_0 + x') \frac{dx'}{dt} - mg \frac{dx}{dt} = 0$$

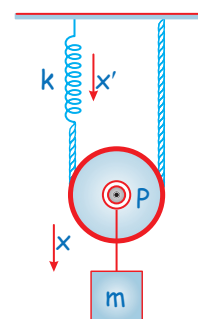
$$mva + k(x_0 + x') \frac{dx'}{dt} - mgv = 0$$

$$mva + k(x_0 + 2x)2v - mgv = 0$$

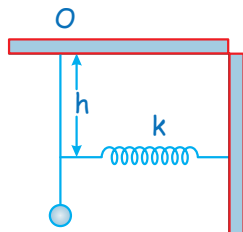
$$ma + 2k(x_0 + 2x) - mg = 0 \quad [\text{As } 2kx_0 = mg, \text{ initially}]$$

$$ma + 4kx = 0$$

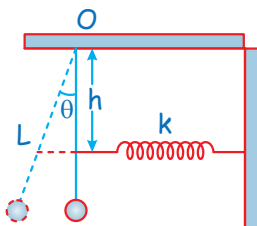
$$a = -\left(\frac{4k}{m}\right)x \Rightarrow \omega = \sqrt{\frac{4k}{m}} \Rightarrow T = 2\pi \sqrt{\frac{m}{4k}}$$



- Q.** A simple pendulum of length L and mass m has a spring of force constant k connected to it at a distance h below its point of suspension (as shown in figure). Find the frequency of vibrations of the system for small values of amplitude.



- Sol.** As shown in figure, if the pendulum is given a small angular displacement θ , the spring will also stretch by $y = h \tan \theta$.



So the restoring torque about O will be due to both force of gravity and elastic force of the spring.

$$\text{i.e. } \tau = -[mg(L \sin \theta) + k(h \tan \theta)h] \quad [\text{Here } F_{\text{spring}} = -k(h \tan \theta)]$$

Now for small θ , $\tan \theta \approx \sin \theta \approx \theta$

$$\text{so, } \tau = -(mgL + kh^2)\theta$$

i.e. restoring torque is linear, so motion is angular SHM.

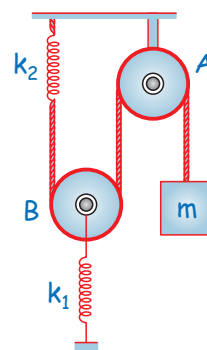
$$\text{Now as } \tau = I\alpha = mL^2 \frac{d^2\theta}{dt^2} \quad (\text{as } I = mL^2)$$

$$\text{So } \frac{d^2\theta}{dt^2} = -\omega^2\theta \quad \text{with } \omega^2 = \left[\frac{mgL + kh^2}{mL^2} \right]$$

This is the standard equation of angular SHM with frequency $f = (\omega/2\pi)$; so here

$$f = \frac{1}{2\pi} \sqrt{\frac{mgL + kh^2}{mL^2}}$$

- Q.** A block of mass m is tied to one end of a string which passes over a smooth fixed pulley A and under a light smooth movable pulley B as shown in figure. The other end of the string is attached to the lower end of a spring of spring constant k_2 . Find the period of small oscillations of mass m about its equilibrium position.



- Sol.** Let F be the extra tension (net restoring force) in the string when mass m is displaced by ' x ' from its mean position. Then, $F = k_2 x_2$ and $2F = k_1 x_1$

where x_1 and x_2 are further (in displaced position) extensions in the springs.

$$\therefore x_1 = \frac{2F}{k_1} \quad \text{and} \quad x_2 = \frac{F}{k_2}$$

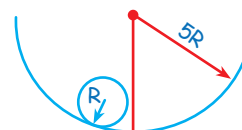
$$\text{Further, } x = 2x_1 + x_2$$

$$= \frac{4F}{k_1} + \frac{F}{k_2} = F \left(\frac{4k_2 + k_1}{k_1 k_2} \right) \quad \text{or } F = \left(\frac{k_1 k_2}{4k_2 + k_1} \right) x$$

As $F \propto x$, motion is simple harmonic in nature.

$$k_{\text{eff}} = \frac{k_1 k_2}{4k_2 + k_1}, \quad T = 2\pi \sqrt{\frac{m(4k_2 + k_1)}{k_1 k_2}}$$

- Q.** A solid sphere (radius = R) rolls without slipping in a cylindrical trough (radius = $5R$). Find the time period of small oscillations.



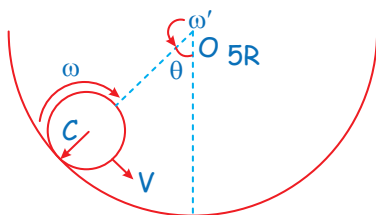
- Sol.** The sphere executes pure rolling in the cylinder. The mean position is at the lowest point in the cylinder. Find the acceleration for small displacement from the mean position and compare with standard equation of SHM to find angular frequency.

$$\text{For pure rolling to take place, } v = R\omega$$

ω = Angular velocity of COM of sphere C about O

$$= \frac{v}{4R} = \frac{R\omega}{4R} = \frac{\omega}{4}$$

$$\therefore \frac{d\omega'}{dt} = \frac{1}{4} \frac{d\omega}{dt} \quad \text{or } \alpha' = \frac{\alpha}{4}, \quad \alpha = \frac{a}{R} \quad \text{for pure rolling;}$$



$$\text{Where, } a = \frac{g \sin \theta}{1 + mR^2} = \frac{5g \sin \theta}{7}$$

$$\text{As, } I = \frac{2}{5}mR^2 \therefore \alpha' = \frac{5g \sin \theta}{28R}$$

For small θ , $\sin \theta \approx \theta$, being restoring in nature,

$$\alpha' = -\frac{5g}{28R} \theta \therefore T = 2\pi \sqrt{\frac{\theta}{\alpha'}} = 2\pi \sqrt{\frac{28R}{5g}}$$

Q. Potential Energy (U) of a body of unit mass moving in one-dimension conservative force field is given by, $U = (x^2 - 4x + 3)$. All units are in S.I.

- Find the equilibrium position of the body.
- Show that oscillations of the body about this equilibrium position are simple harmonic motion & find its time period.
- Find the amplitude of oscillations if speed of the body at equilibrium position is $2\sqrt{6} \text{ m/s}$

Sol. $U = (x^2 - 4x + 3)$

$$(i) F = -\frac{dU}{dx} \Rightarrow F = -2x + 4$$

$$\text{At equilibrium } F = 0$$

$$-2x + 4 = 0 \Rightarrow x = 2\text{m}$$

$$(ii) dF = -2dx \text{ similar to } dF = \frac{\omega^2}{m} dx \text{ (for unit mass) as in SHM}$$

$$2 = \frac{\omega^2}{m} = \omega^2, \omega = \sqrt{2}$$

$$T = \frac{2\pi}{\omega} = \sqrt{2}\pi \text{ sec}$$

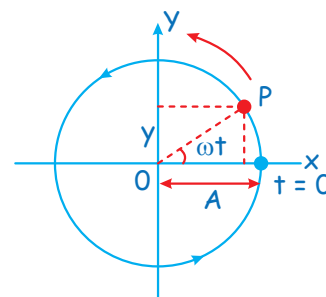
$$(iii) A\omega = 2\sqrt{6}$$

$$A = \frac{2\sqrt{6}}{\sqrt{2}} \Rightarrow A = 2\sqrt{3} \text{ m}$$

GEOMETRICAL INTERPRETATION OF SHM

If a particle is moving with uniform speed along the circumference of a circle, then the straight-line motion of the foot of the perpendicular drawn from the particle on the diameter of the circle is called 'SHM'

Description of SHM Based on Circular Motion



- Draw a circle, having radius equal to amplitude (A) of SHM
- Suppose particle is executing circular motion with angular velocity ω on the circumference of the circle.
- Shadow or projection of particle performs SHM on vertical or horizontal diameter of circle.
- By joining centre of circle to particle position, angle θ is determined from horizontal or vertical diameter. After time t , radius vector will turn by an angle ωt . Based on these unknown values can be calculated.
- As per the position of particle at time t in above diagram, equation of SHM along x-axis can be given as $x = A \cos(\omega t)$ and along y-axis as $y = A \sin(\omega t)$

PHASE AND PHASE DIFFERENCE IN SIMPLE HARMONIC MOTION

Phase: The displacement, velocity and acceleration of a particle executing SHM vary periodically with the angle $\theta = (\omega t + \phi)$ associated with the sine or cosine term. Knowing θ , we can find position as well as its velocity. This angle $(\omega t + \phi)$ is known as the 'phase' of the oscillating particle.

Taking the displacement equation as $x = A \sin(\omega t + \phi)$, we have

at $\omega t + \phi = 0$; $x = 0$ (mean position)

at $\omega t + \phi = \pi/2$; $x = A$ (amplitude position)

at $\omega t + \phi = \pi$; $x = 0$ mean position

at $\omega t + \phi = 3\pi/2$; $x = -A$ and so on

Notice that as time varies indefinitely, phase keeps on changing, 0 to 2π ; because $(\omega t + \phi) = (\omega t + \phi) + 2n\pi$, where n is an integer.

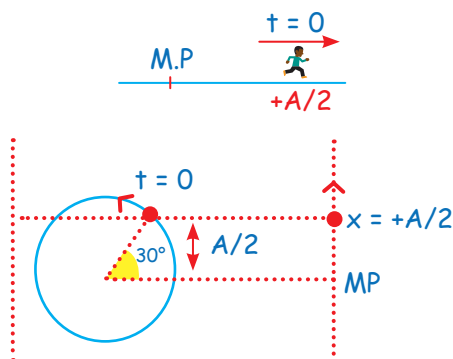
Phase constant (Epoch): The phase of a particle executing SHM, initially is known as the initial phase or phase constant.

Since instantaneous phase = $\omega t + \phi$, so initial phase can be obtained by putting $t = 0$

$\therefore$ Phase constant = ϕ

PHASOR

$$x = A \sin(\omega t + 30^\circ)$$

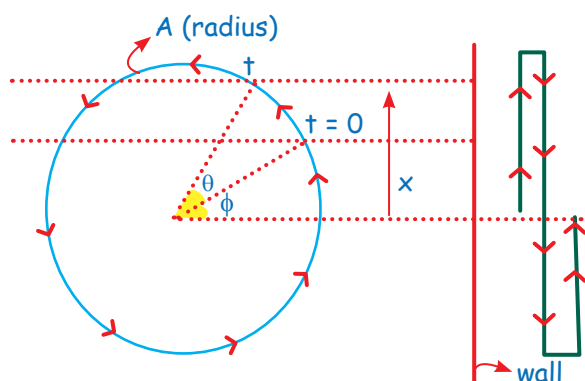


$A \Rightarrow$ uniform circular motion T, ω

shadow $\rightarrow$ SHM. Shadow on wall is SHM.

$$x = A \sin(\theta + \phi)$$

$$x = A \sin(\omega t + \phi)$$



Q. Two particle A and B are performing SHM having same amplitude and same ω such that their eqn of SHM is given as

$$x_1 = A \sin \omega t \rightarrow A$$

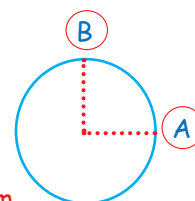
$$x_2 = A \sin(\omega t + 90) \rightarrow B$$

Find

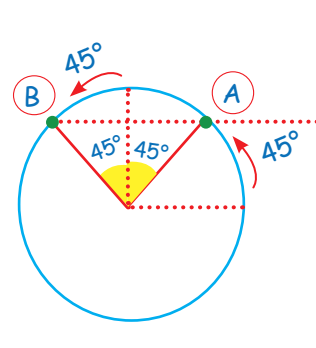
(1) when they will meet

(2) maximum gap between them

(Consider they are not colliding, they just passes through each other)



Sol. (1) When they will meet



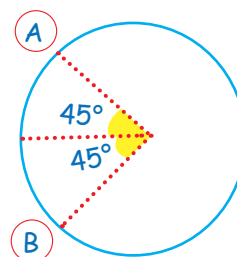
जब दोनों की छाया मिल जाएगी (their projection coincide) अबे ये तो बहुत simple है..... देख phasor मे दोनों के बिच का angle हमेशा same रहेगा (ω same) so, उनके बीच के angle को y-axis के आजू बाजू बराबर बराबर बाट दो और फिर देखो कौनसा particle कितना angle घुमा है



$$\text{They will meet at Time } t = \frac{T}{360^\circ} \times 45^\circ = \frac{T}{8}$$

Max. sepⁿ

In this situation they have maximum gap.



$$(\text{gap})_{\text{max}} = \frac{A}{\sqrt{2}} + \frac{A}{\sqrt{2}} = A\sqrt{2}$$

$$\text{and gap will be max at } t = \frac{T}{360^\circ} \times 135^\circ = \frac{3T}{8}$$

SUPERPOSITION OF SHM

$$\begin{aligned}\vec{F}_1 &= -Kx_1 \rightarrow x_1 = A_1 \sin \omega t \\ \vec{F}_2 &= -Kx_1 \rightarrow x_2 = A_2 \sin (\omega t + \phi) \\ x_{\text{net}} &= A_{\text{net}} \sin (\omega t + \phi) \\ A_{\text{net}} &= \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}\end{aligned}$$

$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 = -K(\vec{x}_1 + \vec{x}_2)$$

$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 = -K\vec{x}_{\text{net}}$$

$$x_1 = A_1 \sin \omega t$$

$$x_2 = A_2 \sin (\omega t + \phi)$$

$$x_{\text{net}} = x_1 + x_2$$

$$x_{\text{net}} = A_{\text{net}} \sin (\omega t + \phi)$$

$$A_{\text{net}} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi} \rightarrow \text{Phase diff.}$$

Q. If $x_1 = A \sin \omega t$

$$x_2 = A \sin (\omega t + 60^\circ)$$

then what will be displacement of superposed SHM.

Sol. $A_{\text{net}} = \sqrt{A^2 + A^2 + 2A.A \cos 60^\circ} = A\sqrt{3}$

$$\tan \theta = \frac{A \sin 60^\circ}{A + A \cos 60^\circ} = \frac{\sqrt{3}/2}{1 + 1/2} = \frac{1}{\sqrt{3}}$$

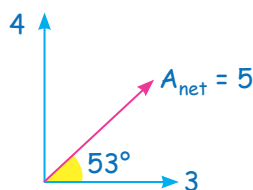
$$\theta = 30^\circ$$

$$x_{\text{net}} = A_{\text{net}} \sin (\omega t + \theta) = A\sqrt{3} \sin (\omega t + 30^\circ)$$

Q. If $x_1 = 3 \sin \omega t$

$$x_2 = 4 \sin (\omega t + 90^\circ), \text{ find } x_{\text{net}} ?$$

Sol. $x_{\text{net}} = 5 \sin (\omega t + 53^\circ)$



Q. Find ' $y_1 + y_2 + y_3$ '

If $y_1 = 10 \sin \omega t$

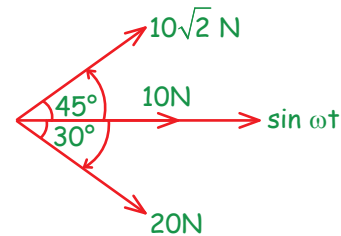
$$y_2 = 10\sqrt{2} \sin (\omega t + 45^\circ)$$

$$y_3 = 20 \sin (\omega t - 30^\circ)$$

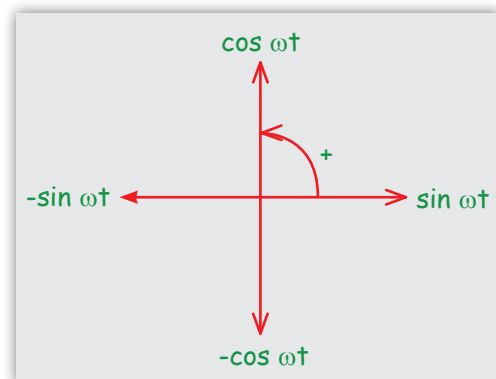
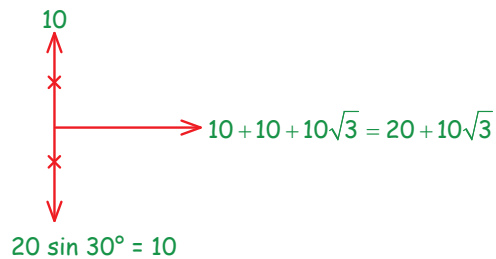
Sol. अब इसे सलीम भईया की आयुषी method से solve करेंगे।

$$\begin{aligned}y_1 &= 10 \sin \omega t \\ y_2 &= 10\sqrt{2} \sin (\omega t + 45^\circ) \\ y_3 &= 20 \sin (\omega t - 30^\circ)\end{aligned}$$

Phase
(+) मतलब ACW



$$y_{\text{net}} = (20 + 10\sqrt{3}) \sin \omega t$$



SKC

आयुषी method 🥰 ये vector नहीं है लेकिन mathematically vector के resultant की तरह से component लेकर solve कर सकते हैं।

सबसे पहले amplitude की value को सही angle के साथ xy-axis पर बनाओ और vector की तरह solve करदो। Finally ये देखो की resultant कितना और किधर आया है।



Same method and same type of question आपको 12th में alternating current में देखने को मिलेगा जैसे

Q. Find $i_1 + i_2 + i_3 + i_4$, if

$$i_1 = 80 \sin \omega t$$

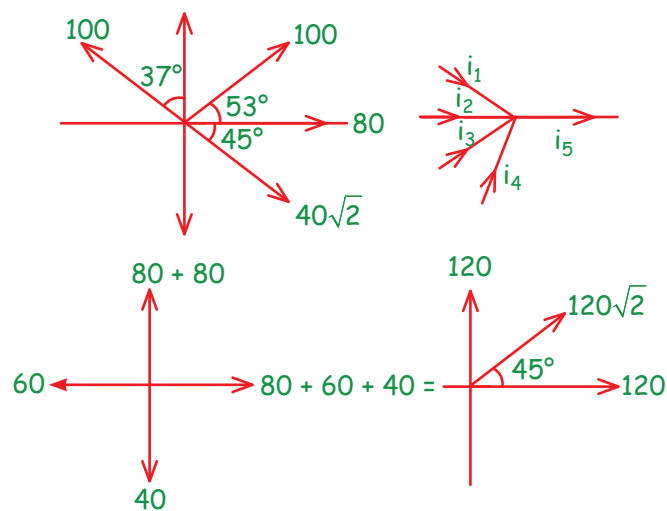
$$i_2 = 40\sqrt{2} \sin(\omega t - 45^\circ)$$

$$i_3 = 100 \sin(\omega t + 53^\circ)$$

$$i_4 = 100 \sin(\omega t + 127^\circ)$$

$$i_{\text{net}} = i_1 + i_2 + i_3 + i_4 = 120\sqrt{2} \sin(\omega t + 45^\circ)$$

Sol. Solve by आयुषी method



Q. If $x_1 = 10 \sin \omega t$

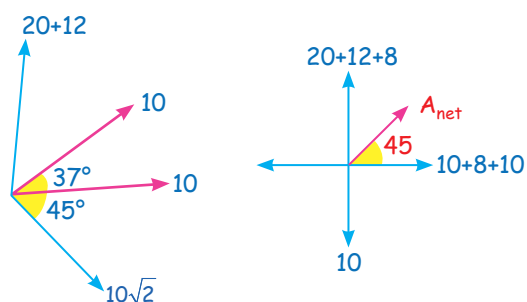
$$x_2 = 20 \sin(\omega t + 90^\circ)$$

$$x_3 = 10 \sin(\omega t + 37^\circ)$$

$$x_4 = 10\sqrt{2} \sin(\omega t - 45^\circ)$$

$$x_5 = 12 \sin(\omega t + 90^\circ), \text{ find } x_{\text{net}} ?$$

Sol. $x_{\text{net}} = 28\sqrt{2} \sin(\omega t + 45^\circ)$



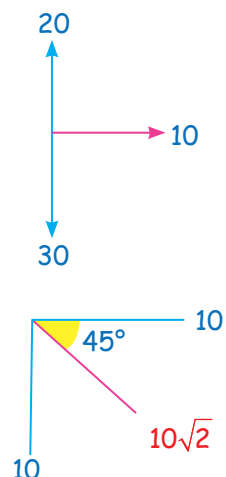
Q. $V_1 = 10 \sin \omega t$

$$V_2 = 20 \sin(\omega t + 90^\circ)$$

$$V_3 = 30 \sin(\omega t - 90^\circ)$$



Sol. $V_1 + V_2 + V_3 = 10\sqrt{2} \sin(\omega t + 45^\circ)$

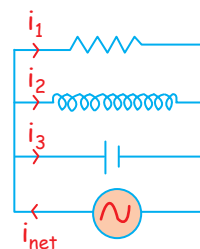


Q. If $i_1 = 10 \sin \omega t$

$$i_2 = 20 \sin(\omega t - 90^\circ)$$

$$i_3 = 30 \sin(\omega t + 90^\circ) \text{ find } i_{\text{net}} ?$$

Sol. $i_{\text{net}} = i_1 + i_2 + i_3 = 10\sqrt{2} \sin(\omega t + 45^\circ)$



अबे देखो तुमने मजे-मजे में alternating current का fundamental question भी कर लिया।

